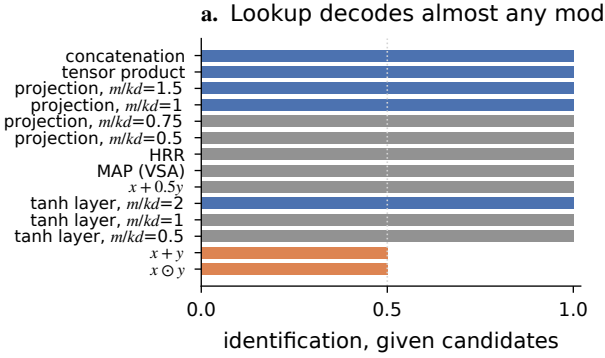
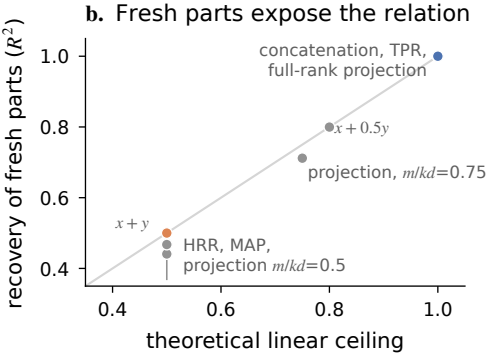
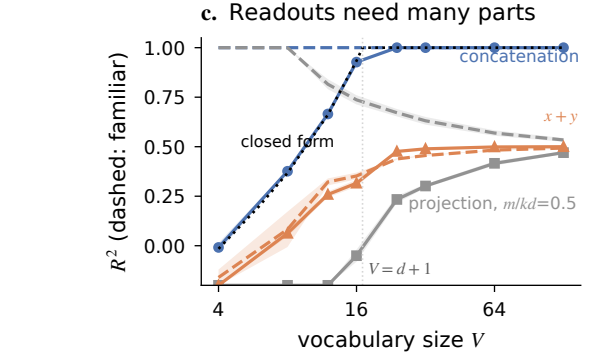
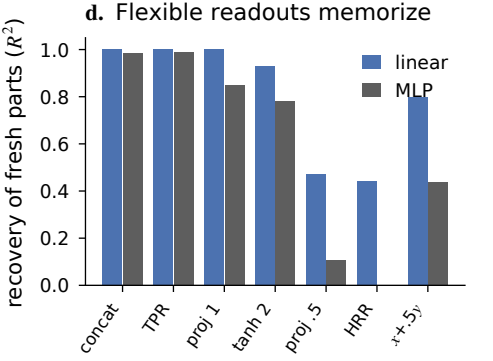

a. Lookup decodes almost any mode
concatenation
tensor product
projection, m/kd=1.5
projection, m/kd=1
projection, m/kd=0.75
projection, m/kd=0.5
HRR
MAP (VSA)
x + 0.5y
tanh layer, m/kd=2
tanh layer, m/kd=1
tanh layer, m/kd=0.5
x + y
x ⊙ y
0.0
0.5
1.0
identification, given candidates


b. Fresh parts expose the relation
recovery of fresh parts (R^2)
1.0
0.8
0.6
0.4
concatenation, TPR,
full-rank projection
x + 0.5y
projection, m/kd=0.75
x + y
HRR, MAP,
projection m/kd=0.5
0.4
0.6
0.8
1.0
theoretical linear ceiling


c. Readouts need many parts
R^2 (dashed: familiar)
1.00
0.75
0.50
0.25
0.00
concatenation
x + y
closed form
projection, m/kd=0.5
V = d + 1
4
16
64
vocabulary size V


d. Flexible readouts memorize
recovery of fresh parts (R^2)
1.0
0.8
0.6
0.4
0.2
0.0
linear
MLP
concat
TPR
proj 1
tanh 2
proj .5
HRR
x+.5y